Process

fork

```
Main()
{
Pid=fork();
If(pid==0)
{
        Child process
        getpid();
        sleep(20);
}
else
{
        Parent process
        getpid();
        sleep();
}
```

Orphan process

```
main()
{
pid=fork();
If(pid==0)
{
        getpid();
        getppid();
        sleep(10);
        getppid();
}
else
{
        getpid();
        getppid();
}
```

Process table

```
[bivasm@cse os]$ ps -l
F S   UID   PID  PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1497 26521 26519  0  80   0 - 27116 wait   pts/2    00:00:00 bash
0 S  1497 27748 26521  0  80   0 -  1624 hrtime pts/2    00:00:00 a.out
1 Z  1497 27749 27748  0  80   0 -     0 exit   pts/2    00:00:00 a.out <defunct>
0 R  1497 27751 26521  3  80   0 - 27032 -      pts/2    00:00:00 ps
[bivasm@cse os]$
```

Zombie

wait()

```
Main()
{
Pid=fork();
If(pid==0)
{
        printf(“First Child process”)

}
else
{
        dip=fork()
        if(dip==0)
        {
                printf(“second child”)
        }
        else
        {
        cpid=wait(0);
        printf(“child died %d”, cpid);
        cpid=wait(0);
        printf(“child died %d”, cpid);
        printf(“Parent”);

}
```

```
Main()
{
Pid=fork();
If(pid==0)
{
        printf("child process")
        exit(i);
}
else
{       wait(&status);
        printf("Parent process");

}
```

Normal termination

Update	0

Abnormal termination

0	Update

```
if (WIFEXITED(status)) {
    printf("Child exited with code %d\n",
WEXITSTATUS(status));
}

if (WIFSIGNALED(status)) {
    printf("Child killed by signal %d\n",
WTERMSIG(status));
}
```

```c
#include <sys/wait.h>
#include <unistd.h>

int main() {
    int status;
    pid_t pid = fork();

    if (pid == 0) {
        // Child
        printf("Child running\n");
        exit(3);
    } else {
        // Parent
        wait(&status);

        if (WIFEXITED(status)) {
            printf("Child exited normally with code %d\n",
                   WEXITSTATUS(status));
        }
    }
    return 0;
}
```

waitpid

```c
pid_t waitpid(pid_t pid, int *statusPtr, int options);
```

```c
int main (){
    int pid;
    int status;

    printf("Parent: %d\n", getpid());

    pid = fork();
    if (pid == 0){
        printf("Child %d\n", getpid());
        sleep(2);
        exit(EXIT_SUCCESS);
    }

//Comment from here to...
    //Parent waits process pid (child)
    waitpid(pid, &status, 0);
    //Option is 0 since I check it later
```

exec

./Ex1

```
Main()
{
        printf("before");
        execl("usr/guest/ex2", "ex2", (char*)0);
        printf("after");
}
```

exec

Ex1

```
main(int argc, char* argv[])
{
        printf("before");
        execl(argv[1],argv[2], argv[3], argv[4], (char*)0);
        printf("after");
}
```

./Ex1 /usr/guest/ex2 ex2 hello world

Ex2

```
main(int argc, char* argv[])
{
        printf("%s %s %s", argv[0], argv[1], argv[2]);
}
```

exec

./Ex1 /bin/ls ls -l

Ex1

```
main(int argc, char* argv[])
{
        printf(“before”);
        execl(argv[1],argv[2], argv[3], argv[4], (char*)0);
        printf(“after”);
}
```

exec

Execv(path, temp)

Execvp(file, temp)

Temp[0]="ex2"

Temp[1]="hello"

Temp[2]="world"

Temp[3]='\0'

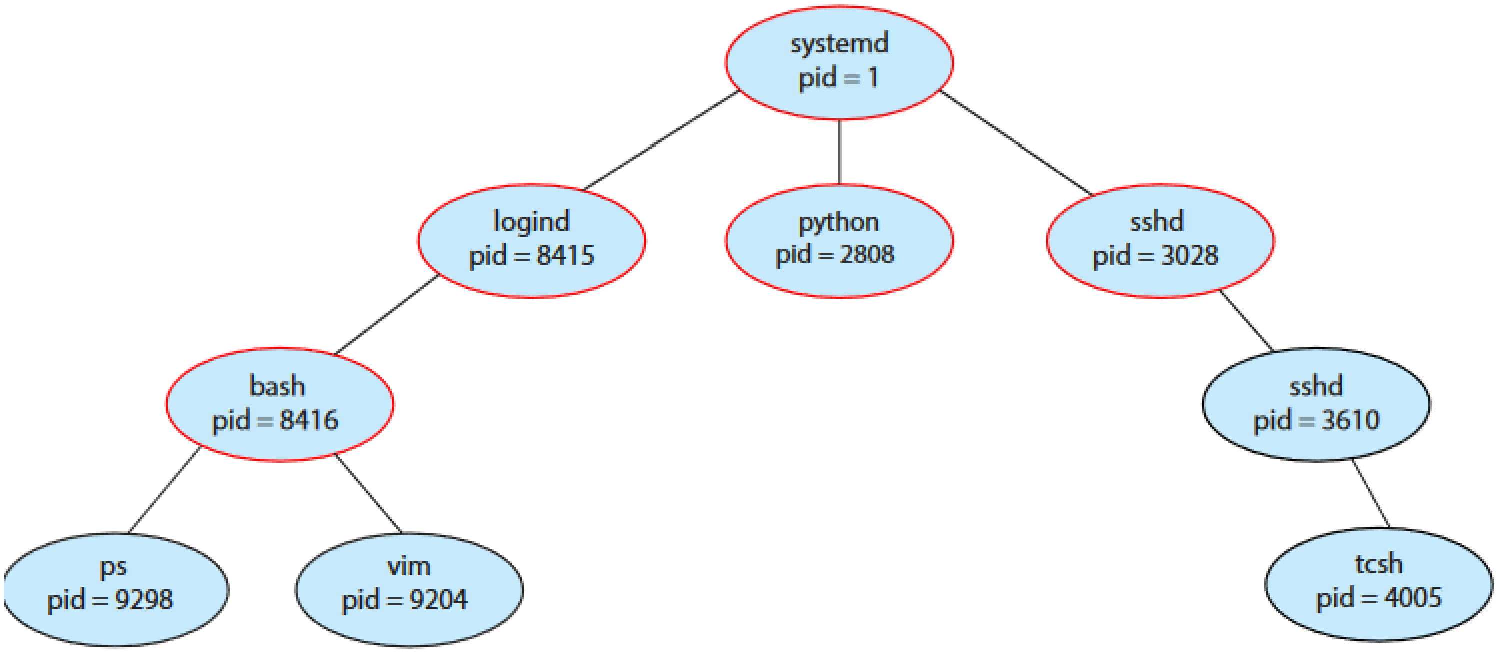
Execvp(temp[0], temp)

Ex2

Printf(argv[0], argv[1], argv[2])

Ex2 hello world

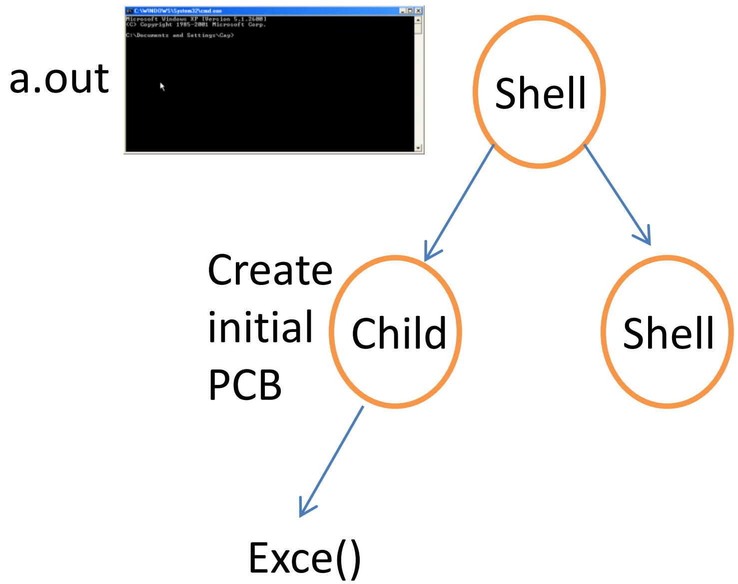
shell



systemd
pid = 1
logind
pid = 8415
python
pid = 2808
sshd
pid = 3028
bash
pid = 8416
sshd
pid = 3610
ps
pid = 9298
vim
pid = 9204
tcsh
pid = 4005

```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>

int main()
{
pid_t pid;

    /* fork a child process */
    pid = fork();

    if (pid < 0) { /* error occurred */
       fprintf(stderr, "Fork Failed");
       return 1;
    }
    else if (pid == 0) { /* child process */
       execlp("/bin/ls","ls",NULL);
    }
    else { /* parent process */
       /* parent will wait for the child to complete */
       wait(NULL);
       printf("Child Complete");
    }

    return 0;
}
```

shell



shell

```
int main()
{
        int pid;
        while(1)
        {
                pid=fork()
                if (pid==0)
                {
                        exec(“ls”);
                }
                else
                {
                        wait(NULL);
                }
        }

}
```